

Wild cubic inverse-image towers in characteristic three

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Abstract

Let k be any field of characteristic three and let $a \in k^*$. We determine the splitting fields of $(X^3 + aX^2)^{\circ n} - t$ over $k(t)$ for every inverse-image height $n \geq 1$, with t transcendental. Their arithmetic and geometric Galois groups coincide with the classical signature-constrained ternary groups E_n , and have order $2^{3^n-1}3^{(3^n-1)/2}$. The proof separates two different kinds of independence. At each step the only relations among the relevant square classes are the sibling relations, and all 3^n resulting Artin–Schreier classes are globally independent. In contrast, after extending the constants to an algebraic closure, these classes span only one dimension at every completion above infinity. A simultaneous induction on the tree group and this local rank gives the complete geometric branch data: the ramification indices at zero and infinity are 2^n and $2 \cdot 3^n$, and the corresponding different exponents are $2^n - 1$ and $3^{n+1} - 2$. The geometric genus is $1 + \frac{|E_n|}{2}(\frac{1}{2} - 2^{-n} - 3^{-n})$, giving genus 46 at height two. The result is a wild characteristic-three realization with ramification data, not a new construction of the abstract groups E_n . It concerns generic inverse images, not forward periodic points or finite-field extension degrees.

1 Introduction

An iterated splitting field can have many independent global generators while acquiring very little ramification at a particular place. For the cubic

$$f(X) = X^3 + aX^2 \quad (\text{char } k = 3, a \neq 0),$$

this distinction determines the entire generic inverse-image tower. The finite critical point is fixed, and infinity is wildly ramified. At height n , a quadratic radical step produces 3^n Artin–Schreier equations. Their classes are independent over the global function field, but become proportional over each geometric completion at infinity. Tracking these two ranks simultaneously yields both the Galois group and the ramification at every height.

The groups that occur here are already known. Benedetto, Faber, Hutz, Juul and Yasufuku introduced the signature-constrained ternary groups E_n , computed their orders, and realized them for the cubic $-2X^3 + 3X^2$ in a number-field setting; their geometric realization is given in Corollary 1.3 of [2]. Theorem 2.3.1 of Bouw–Ejder–Karemaker [3] places these groups in the wider theory of normalized dynamical Belyi maps. Ejder [4] uses the same local product-sign description in studying arithmetic monodromy. Thus neither E_n nor the formula for its order is a new group-theoretic object in this paper.

Those characteristic-zero Belyi realizations do not supply the wild local calculation required here. Our map has the two geometric branch values 0 and ∞ , with ramification divisible by the characteristic at infinity. The elementary square-root and Artin–Schreier normal form retains all constants in a , so the result does not require a square root of a or an initial extension of the base field.

This also makes it possible to descend the geometric group calculation to every characteristic-three field.

Recent general monodromy results have different characteristic hypotheses. The version of Adams–Hyde [1] cited here treats unicritical power polynomials of degree prime to the characteristic. The cited version of Hlushchanka–Lukina–Wardell [5] treats a number-field cubic setting and discusses extension to characteristics prime to two and three. Neither stated scope includes the present characteristic-three family. These comparisons concern the specified sources and versions; they are not an assertion of exhaustive literature priority.

The main results answer one question: what are the generic splitting fields of $f^{\circ n}(X) - t$, together with their geometric branch data, for all n ? Three parts of the answer are useful to distinguish. First, a single compatible labelling of the inverse-image tree gives arithmetic and geometric group E_n over an arbitrary base field. Second, over algebraically closed constants, the square-class relations and the two Artin–Schreier ranks are determined exactly. Third, the resulting ramification indices give explicit different exponents and genera. The substantive calculation is the simultaneous global/local induction, not a list of orders or low-height examples.

Section 2 states the results and their precise base-field conventions. Section 3 records the cubic normal form and the classical local facts needed in the proof. Section 4 establishes the tree-group upper bound before any maximality assertion is used. Sections 5 and 6 prove the mixed-radical induction, including descent to the original constants. Section 7 computes the different and genus.

2 The tower and the main statements

Fix a field k of characteristic three, an element $a \in k^*$ and a transcendental element t . All splitting fields are taken in a fixed algebraic closure of $k(t)$. Set

$$f(X) = X^3 + aX^2, \quad L_n = k(t)(f^{-n}(t)), \quad G_n = \text{Gal}(L_n/k(t)), \quad n \geq 1.$$

The notation $f^{\circ n}$ denotes composition, and $f^{-n}(t)$ is the set of roots of $f^{\circ n}(X) - t$. The field L_n is the splitting field, not merely the field generated by one chosen root. The polynomials are separable and irreducible; this will also follow directly in Section 3.

The root t , its three preimages, and their successive preimages form the regular rooted ternary tree T_∞ . Its truncation at height n is T_n . A labelling assigns 1, 2, 3 to the children of every vertex; compatibility means that the labellings at finite heights are restrictions of this one labelling. Put $W_n = \text{Aut}(T_n)$. If $g = (g_1, g_2, g_3)\sigma \in W_n$ and $n \geq 2$, define

$$\text{sgn}_2(g) = \text{sgn}(\sigma) \prod_{i=1}^3 \text{sgn}(g_i|_{T_1}).$$

Here the signs refer to permutations of three children. Following the classical definition in [2, Section 2] and [4, Definition 2.2], set

$$E_1 = \mathfrak{S}_3, \quad E_n = (E_{n-1}^3 \rtimes \mathfrak{S}_3) \cap \ker \text{sgn}_2 \quad (n \geq 2). \quad (1)$$

Equivalently, at every vertex having two levels below it, its local sign equals the product of the local signs at its three children. The restriction maps define $E = \varprojlim_n E_n$. Equality with these groups below is equality in the labelled tree action, not just an abstract isomorphism or a condition on the sign of all leaves.

Choose an algebraic closure $\bar{k}$ of k in the ambient algebraic closure and put $\bar{L}_n = \bar{k}L_n$. Its group over $\bar{k}(t)$ is the geometric group. All local assertions in the next two geometric statements concern $\bar{L}_n/\bar{k}(t)$.

Theorem 2.1 (Arithmetic and geometric tower). *For every k and a as above there is one compatible labelling such that, for all $n \geq 1$,*

$$\mathrm{Gal}(L_n/k(t)) = \mathrm{Gal}(\overline{L}_n/\overline{k}(t)) = E_n, \quad |E_n| = 2^{3^{n-1}} 3^{(3^n-1)/2}. \quad (2)$$

The extension L_n/k is regular; in particular its algebraic constant field is exactly k . The arithmetic and geometric inverse-limit actions both equal E in the same labelling.

For the radical description, let $\mathcal{A}_n = f^{-n}(t)$ and, for each $\alpha \in \mathcal{A}_n$, choose β_α with $\beta_\alpha^2 = \alpha/a^3$. Define

$$\overline{M}_n = \overline{L}_n(\beta_\alpha : \alpha \in \mathcal{A}_n), \quad \wp(u) = u^3 - u.$$

The choice of each square-root sign does not affect the fields or the ranks in the following statement. A *sibling triple* is the set of three roots α with the same parent $f(\alpha)$.

Theorem 2.2 (Global and local radical ranks). *For every $n \geq 1$, the relation space among the square classes α/a^3 in $\overline{L}_n^*/\overline{L}_n^{*2}$ is generated by the indicator vectors of the sibling triples. Consequently*

$$[\overline{M}_n : \overline{L}_n] = 2^{2 \cdot 3^{n-1}}, \quad [\overline{L}_{n+1} : \overline{M}_n] = 3^{3^n}. \quad (3)$$

The 3^n classes $[\beta_\alpha]$ in $\overline{M}_n/\wp(\overline{M}_n)$ are linearly independent over $\mathbb{F}_3$. The extension $\overline{M}_n/\overline{L}_n$ splits completely at every place above infinity. At every corresponding completion of $\overline{M}_n$, the images of those same Artin–Schreier classes span exactly one dimension over $\mathbb{F}_3$.

Let $\overline{C}_n$ be the smooth projective curve over $\overline{k}$ with function field $\overline{L}_n$. All valuations in the next theorem are normalized to have value group $\mathbb{Z}$ in the field upstairs; a different exponent is measured in that normalization.

Theorem 2.3 (Geometric branch data and genus). *The cover $\overline{C}_n \rightarrow \mathbb{P}_k^1$ has branch set exactly $\{0, \infty\}$. At every point above zero, its ramification index and different exponent are*

$$e_{0,n} = 2^n, \quad d_{0,n} = 2^n - 1. \quad (4)$$

At every point above infinity they are

$$e_{\infty,n} = 2 \cdot 3^n, \quad d_{\infty,n} = 3^{n+1} - 2. \quad (5)$$

Its genus is

$$g_n = 1 + \frac{|E_n|}{2} \left(\frac{1}{2} - \frac{1}{2^n} - \frac{1}{3^n} \right). \quad (6)$$

In particular, $g_1 = 0$ and $g_2 = 46$.

Theorem 2.1 is over an arbitrary, possibly imperfect, constant field. Theorems 2.2 and 2.3 are geometric statements after adjoining $\overline{k}$; they do not assert that all branch points over the original k have residue degree one. The height n is neither a forward-period clock nor a finite-field extension degree. No surjectivity statement for a specialization $t = t_0 \in k$ is included.

The proof proceeds simultaneously in the group and the infinity index. The field-independent signature bound is proved first. Assuming the result at height n , the zero-place valuations determine the next square-class rank, while the already known infinity index gives pole order one for the square roots. Character projections prove their global independence, and a separate leading-coefficient calculation proves local rank one. The two conclusions then establish the next group and the next infinity index. This order avoids using the unproved group at height $n+1$ in the rank calculation.

3 The cubic normal form and local tools

We first record the explicit field description used at every vertex. Its constants matter because the arithmetic conclusion is over the original field k .

Lemma 3.1 (Cubic splitting field). *Let F be a field of characteristic three containing $a \neq 0$ and $\alpha \neq 0$. Choose β, z in an algebraic closure with*

$$\beta^2 = \alpha/a^3, \quad z^3 - z = \beta.$$

Then the splitting field of $f(X) - \alpha$ over F is $F(\beta, z)$. Its three roots are

$$a(z^2 - 1), \quad a(z^2 + z), \quad a(z^2 - z). \quad (7)$$

Moreover,

$$\text{disc}_X(f(X) - \alpha) = a^3\alpha, \quad \prod_{f(\gamma)=\alpha} \gamma = \alpha. \quad (8)$$

Proof. Direct expansion in characteristic three gives

$$f(a(z^2 - 1)) = a^3(z^3 - z)^2. \quad (9)$$

The three substitutions $z, z + 1, z - 1$ give exactly the expressions in (7). They are distinct because $f(X) - \alpha$ is separable: its derivative is $-aX$, and $X = 0$ is not a root. Thus its splitting field is contained in $F(\beta, z)$. Conversely, subtracting the second expression in (7) from the third gives az . Hence the field generated by the three roots contains z and then $\beta = z^3 - z$. This proves equality of fields, not just an embedding into a radical extension.

The product formula follows from the constant coefficient $-\alpha$. For the discriminant, the resultant with the derivative is $(-a)^3 \prod \gamma = -a^3\alpha$. The degree-three discriminant sign is negative, giving the first formula in (8). $\square$

In particular, if Δ is a Vandermonde product of the three roots, then

$$(\Delta/a^3)^2 = \alpha/a^3. \quad (10)$$

Thus the quadratic radical is already in the cubic splitting field, with no need to adjoin a square root of a .

Lemma 3.2 (Generic roots and the first height). *For every $n \geq 1$, the polynomial $f^{\circ n}(X) - t$ is separable and irreducible over $k(t)$. At height one,*

$$L_1 = k(z), \quad t = a^3(z^3 - z)^2,$$

and its Galois group is the group of six transformations $z \mapsto \varepsilon z + j$, with $\varepsilon \in \{1, -1\}$ and $j \in \mathbb{F}_3$. In particular $G_1 = \mathfrak{S}_3$, L_1/k is regular, and the geometric ramification index at infinity is six.

Proof. The polynomial $f^{\circ n}(X) - t$ is irreducible in $k[X, t]$: its quotient ring is $k[X]$ under $t = f^{\circ n}(X)$. It is primitive as a polynomial in X over $k[t]$, so Gauss's lemma gives irreducibility over $k(t)$. The chain rule gives

$$(f^{\circ n})'(X) = (-a)^n \prod_{j=0}^{n-1} f^{\circ j}(X). \quad (11)$$

If a zero of this derivative also satisfied $f^{\text{on}}(X) = t$, some iterate would be zero and all later iterates would be zero, contradicting the transcendence of t . This proves separability.

For $n = 1$, Lemma 3.1 identifies the splitting field with $k(z)$ and the displayed parameter t . This rational function has degree six, so $[k(z) : k(t)] = 6$. Each of the six indicated transformations fixes t , and they form the full Galois group, isomorphic to $\mathfrak{S}_3$. The field $k(z)$ is a rational function field over k , so it has no additional constants and is regular. The unique pole of t has order six; this gives the geometric infinity index. $\square$

For reference, we give the radical-degree fact used below. This also specifies precisely what is meant by the rank of a collection of classes.

Lemma 3.3 (Classical radical degrees). *Let F be a field of characteristic different from two. For $b_1, \dots, b_s \in F^*$, the degree of $F(\sqrt{b_1}, \dots, \sqrt{b_s})/F$ is 2^r , where r is the dimension of the span of $[b_1], \dots, [b_s]$ in F^*/F^{*2} . In characteristic three, if $z_i^3 - z_i = c_i \in F$, then $[F(z_1, \dots, z_s) : F] = 3^r$, where r is the dimension of the span of $[c_i]$ in $F/\wp(F)$ over $\mathbb{F}_3$.*

Proof. In the quadratic case the compositum is Galois, with group a subgroup $H \subseteq \mathbb{F}_2^s$ recording the signs of the square roots. An exponent vector e annihilates H precisely when $\prod_i \sqrt{b_i}^{e_i}$ is fixed by H , hence lies in F . This is equivalent to $\prod_i b_i^{e_i}$ being a square in F ; the converse uses that the two square roots of a square are in F . The annihilator of H is therefore the relation space of the square classes. The perfect coordinate pairing on $\mathbb{F}_2^s$ gives $\dim H = r$.

For Artin–Schreier equations, the compositum is again Galois, now with $H \subseteq \mathbb{F}_3^s$ recording the translations of the z_i . A vector e annihilates H if and only if $\sum_i e_i z_i \in F$. Such a vector gives $\sum_i e_i c_i \in \wp(F)$. Conversely, if $\sum_i e_i c_i = u^3 - u$ for $u \in F$, then $\sum_i e_i z_i - u$ is a root of $X^3 - X$ and belongs to $\mathbb{F}_3$. Thus e annihilates H . The same pairing argument proves the claim. $\square$

All completions used in the geometric calculation are one-variable complete discretely valued fields with algebraically closed constant and residue field. The following local facts will be applied in that setting.

Lemma 3.4 (Geometric local facts). *Let $K = \kappa((s))$, where κ is algebraically closed of characteristic three, and let $v_K(s) = 1$.*

1. *Every unit of $\kappa[[s]]$ is a square in K . Every element of $\kappa[[s]]$ belongs to $\wp(K)$.*
2. *If $v_K(b) = -1$, then $b \notin \wp(K)$, and the extension defined by $z^3 - z = b$ is totally ramified of degree three.*
3. *The compositum of finitely many tame extensions of K , of ramification indices $e_1, \dots, e_r$, is tame of index $\text{lcm}(e_1, \dots, e_r)$.*

Proof. For the first statement, solve the square-root or Artin–Schreier equation in the algebraically closed residue field. Hensel’s lemma lifts the root: the derivatives are respectively a nonzero multiple of the residue square root and -1 .

If u has a pole, $v_K(u^3 - u) = 3v_K(u)$; if u is integral, $u^3 - u$ has no pole. Thus an element of pole order one is not an Artin–Schreier image. The equation gives a degree-three extension. In that extension, z must have negative valuation and $3v(z) = -e$, where e is the ramification index over K . Hence $3 \mid e$, and the degree bound forces $e = 3$.

For the last statement the residue extension is trivial. A tame extension of index e can be written $K(s^{1/e})$: for a uniformizer π upstairs, write $s = u\pi^e$, take an e th root of the unit u by Hensel’s lemma, and adjust π . Its degree is e , and κ already contains the e th roots of unity. The asserted compositum and its index follow. $\square$

We also use the standard local different formula: if a finite separable geometric extension is written $\kappa((s)) \subseteq \kappa((u))$ with $s = H(u)$, its different exponent is $\text{ord}_u H'(u)$. This follows from the presentation of relative differentials, generated by du with relation $H'(u)du = 0$, or equivalently from the monogenic different formula. For a tower the exponents satisfy

$$d(L/K) = d(L/F) + e(L/F)d(F/K). \quad (12)$$

These local facts, Kummer and Artin–Schreier theory, and the Riemann–Hurwitz formula are classical function-field tools; see, for example, [6, Chapter 3].

4 A compatible signature bound

The discriminant relations must hold simultaneously on the actual tree. A square discriminant at one height, without a compatible choice of signs at all vertices, would not prove the required bound.

Proposition 4.1 (Signature upper bound). *Over the original field k , the inverse-image tree admits a compatible labelling for which $G_n \subseteq E_n$ for every $n \geq 1$.*

Proof. For a vertex v with value x_v , order its three children and let Δ_v be their Vandermonde product. By (8),

$$\Delta_v^2 = a^3 x_v, \quad \left(\prod_{w \text{ child of } v} \Delta_w \right)^2 = a^9 x_v.$$

All these quantities are nonzero because the generic roots are distinct and no vertex value is zero. Consequently

$$\frac{\prod_{w \text{ child of } v} \Delta_w}{a^3 \Delta_v} \in \{1, -1\}.$$

Choose the order of the first-level children. Having chosen all orders through one height, choose orders one height lower so that

$$\prod_{w \text{ child of } v} \Delta_w = a^3 \Delta_v \quad (13)$$

at every newly treated parent v . If the sign is initially wrong, an odd permutation of the children of one child w changes exactly one factor on the left. The correction is independent for different parents. It does not alter a Vandermonde already fixed higher in the tree, so it preserves the previously imposed relations. Induction produces one labelling of the infinite tree.

For $g \in G_n$, let $s_v(g)$ be the sign of its permutation of the three ordered children of v onto those of $g(v)$. Apply g to (13) and compare it with that identity at $g(v)$. Since a is fixed, the comparison gives

$$s_v(g) = \prod_{w \text{ child of } v} s_w(g).$$

These are exactly the local defining relations of E_n at all vertices with two remaining levels. They prove $G_n \subseteq E_n$. The construction took place over k itself, without adjoining constants. $\square$

We record two elementary consequences of the classical group definition. The order formula is [2, Proposition 2.2]; its short recurrence is included to make the later degree comparison explicit.

Lemma 4.2 (Order and bottom actions). *The groups in (1) satisfy*

$$|E_1| = 6, \quad |E_{n+1}| = 3|E_n|^3, \quad \frac{|E_{n+1}|}{|E_n|} = 2^{2 \cdot 3^{n-1}} 3^{3^n}. \quad (14)$$

For every bottom parent of T_n , E_n contains the subgroup $\mathfrak{A}_3$ permuting that parent's three children and fixing every other vertex.

Proof. The product-sign homomorphism on $E_n^3 \rtimes \mathfrak{S}_3$ is onto: a root transposition with identity sections has value -1 . Its kernel therefore has order $6|E_n|^3/2 = 3|E_n|^3$. Solving this recurrence gives

$$|E_n| = 2^{3^{n-1}} 3^{(3^n-1)/2},$$

and division gives the final expression in (14). An independent bottom three-cycle has positive local sign everywhere, so it satisfies every local product-sign relation. The same holds for its square and the identity. $\square$

The bottom actions in Lemma 4.2 will be used only after the induction has proved $G_n = E_n$ at the current height. They will not be assumed for G_{n+1} .

5 Zero-place valuations and the square-class rank

In this section and until the descent in Section 6.5, we work over $\bar{k}$. We retain the bars in $\bar{L}_n$ and $\bar{M}_n$ to make that convention visible. The zero-place analysis does not assume maximality of the next Galois group.

Lemma 5.1 (Branch support and the tame place). *The extension $\bar{L}_n/\bar{k}(t)$ is unramified away from zero and infinity. Its ramification index at every place above zero is 2^n . If P is such a place, with normalized valuation v_P , then every $\alpha \in \mathcal{A}_n$ is integral, and*

$$v_P(\alpha) = \begin{cases} 1, & \alpha \text{ specializes to } 0, \\ 0, & \alpha \text{ specializes to a nonzero element of } \bar{k}. \end{cases} \quad (15)$$

Exactly 2^n roots specialize to zero. In every bottom sibling triple their valuation-parity vector is either 000 or one of 110, 101, 011.

Proof. Equation (11) shows that every finite critical value of the root cover $t = f^{\circ n}(X)$ is zero: a vanishing intermediate iterate remains zero thereafter. Thus that cover is unramified outside zero and infinity. The same is true of its Galois closure, since conjugates and composita of unramified local extensions remain unramified.

Above zero, f has local degree two at 0 and local degree one at its other preimage $-a$. The ramification index of a point in the n th root cover is the product of these local degrees along its orbit, hence a power of two dividing 2^n . At $X = 0$ it is exactly 2^n . These local extensions are tame. By Lemma 3.4, their local Galois closure has index the least common multiple of the root-cover indices, which is 2^n . Consequently $v_P(t) = 2^n$.

Every root is integral at P , by monicity. The reduction of $f^{\circ n}(X) - t$ is $f^{\circ n}(X)$; its zero root has multiplicity 2^n . Since the polynomial splits into linear factors upstairs, precisely that many roots specialize to zero. Near zero, $f^{\circ n}(X)$ has a nonzero leading term of degree 2^n . Thus if α specializes to zero,

$$2^n v_P(\alpha) = v_P(f^{\circ n}(\alpha)) = v_P(t) = 2^n,$$

which gives $v_P(\alpha) = 1$. All other roots are units.

At a vertex specializing to zero, the reduced child equation is $X^2(X + a) = 0$, so two children specialize to zero and one to $-a$. A vertex specializing to a nonzero value has no zero child. The zero-specializing vertices therefore form a binary subtree, and the asserted parity alternatives follow. $\square$

Proposition 5.2 (Exact square-class relations). *Assume $\text{Gal}(\overline{L}_n/\overline{k}(t)) = E_n$ in the compatible labelling. Then the square-class assertion of Theorem 2.2 holds at height n , and*

$$[\overline{M}_n : \overline{L}_n] = 2^{2 \cdot 3^{n-1}}.$$

Every class $[\alpha/a^3]$ is nonzero, and the classes belonging to distinct roots are distinct.

Proof. Let $r_n = 3^{n-1}$ be the number of bottom sibling triples. Inside $\mathbb{F}_2^{A_n}$ define

$$V_n = \bigoplus_{\text{sibling triples}} \{(u_1, u_2, u_3) \in \mathbb{F}_2^3 : u_1 + u_2 + u_3 = 0\}.$$

It has dimension $2r_n$. By Lemma 5.1, the parity vectors of root valuations at places over zero all lie in V_n . We claim that their span is all of V_n .

Choose a place and a bottom triple on which its parity vector is nonzero. Such a triple exists because 2^n roots specialize to zero. The induction hypothesis and Lemma 4.2 give a Galois element acting as a three-cycle on that triple and fixing all other triples. Conjugating the place permutes its valuation vector accordingly. Adding the original and conjugated vectors over $\mathbb{F}_2$ cancels every other triple and leaves a nonzero even vector on the chosen triple. Its three cyclic translates span the two-dimensional even subspace on that triple. The Galois group acts transitively on bottom parents, since $f^{o(n-1)}(X) - t$ is irreducible; for $n = 1$ there is just the parent t . Conjugation therefore gives the even subspace on every triple. This proves the claim.

If

$$\prod_{\alpha \in A_n} (\alpha/a^3)^{e_\alpha} \in \overline{L}_n^{*2}, \quad e_\alpha \in \mathbb{F}_2,$$

its valuation is even at every place over zero. The exponent vector e is consequently orthogonal to V_n under the coordinate pairing. The orthogonal complement $V_n^\perp$ is the span of the all-one vectors on the disjoint sibling triples. It remains to verify that these necessary relations actually hold.

For a sibling triple with parent η , formula (8) gives

$$\prod_{f(\alpha)=\eta} \frac{\alpha}{a^3} = \frac{\eta}{a^9} = \left(\frac{\beta_\eta}{a^3} \right)^2. \quad (16)$$

Here one may choose $\beta_\eta = \Delta_\eta/a^3 \in \overline{L}_n$ by (10), because $f(X) - \eta$ has split at height n ; it satisfies $\beta_\eta^2 = \eta/a^3$. This includes $n = 1$, when $\eta = t$. Thus every sibling vector is a genuine relation. There are no others.

The square classes have rank $3^n - r_n = 2r_n$, so Lemma 3.3 gives the degree assertion. A nonzero vector in the span of disjoint sibling indicators has weight at least three, and its weight is a multiple of three. Neither a weight-one vector nor a weight-two vector can be a relation. The individual classes are therefore nonzero and pairwise distinct. $\square$

This proposition uses only the group at height n . The parity vectors come from the tame zero-place geometry, and the relations are verified by a product identity in the already known splitting field. No degree or independence statement at height $n + 1$ has entered its proof.

6 The local and global Artin–Schreier induction

We now carry two induction hypotheses at height n :

$$\text{Gal}(\overline{L}_n/\overline{k}(t)) = E_n, \quad e_\infty(\overline{L}_n/\overline{k}(t)) = 2 \cdot 3^n. \quad (17)$$

Both hold at height one by Lemma 3.2. The second hypothesis controls valuations before the next global degree is known.

6.1 Kummer splitting at infinity

Lemma 6.1 (Pole order after the quadratic step). *Assume (17). The extension $\overline{M}_n/\overline{L}_n$ splits completely at every place above infinity. At each such normalized place of $\overline{M}_n$,*

$$v(\alpha) = -2, \quad v(\beta_\alpha) = -1 \quad \text{for every } \alpha \in \mathcal{A}_n. \quad (18)$$

In particular, every $[\beta_\alpha]$ is nonzero in $\overline{M}_n/\wp(\overline{M}_n)$.

Proof. Let Q be a normalized place of $\overline{L}_n$ over infinity. The second hypothesis gives $v_Q(t) = -2 \cdot 3^n$. A root α cannot be integral, since $f^{\circ n}(\alpha) = t$. The polynomial $f^{\circ n}$ is monic of degree 3^n , with constant-field coefficients. Its leading term therefore has strictly smallest valuation at α , giving

$$3^n v_Q(\alpha) = v_Q(t) = -2 \cdot 3^n.$$

Thus $v_Q(\alpha) = -2$.

Each α/a^3 is a unit times an even power of a uniformizer. Every such unit has a square root in the geometric completion by Lemma 3.4. All the quadratic polynomials used to define $\overline{M}_n$ already split in that completion. Since $\overline{M}_n/\overline{L}_n$ is Galois, it splits completely there. The normalized valuation is consequently unchanged after the quadratic step, and $v(\beta_\alpha) = -1$.

An element with pole order one cannot be an Artin–Schreier image in the completion, by Lemma 3.4. A global equation $\beta_\alpha = u^3 - u$ would remain such an equation in every completion, which is impossible. Hence its global class is nonzero. $\square$

6.2 Independent global classes

The nonvanishing in Lemma 6.1 is only an individual statement. To pass from nonvanishing to independence, we use the distinct characters produced by the exact Kummer relations.

Proposition 6.2 (Global Artin–Schreier rank). *Under (17), the 3^n elements $[\beta_\alpha] \in \overline{M}_n/\wp(\overline{M}_n)$ are linearly independent over $\mathbb{F}_3$. Moreover,*

$$\overline{L}_{n+1} = \overline{M}_n(z_\alpha : \alpha \in \mathcal{A}_n), \quad z_\alpha^3 - z_\alpha = \beta_\alpha, \quad (19)$$

and $[\overline{L}_{n+1} : \overline{M}_n] = 3^{3^n}$.

Proof. Let $H = \text{Gal}(\overline{M}_n/\overline{L}_n)$. By Proposition 5.2, this is an elementary abelian two-group. The square-root line for each α is a character line:

$$h(\beta_\alpha) = \chi_\alpha(h)\beta_\alpha, \quad \chi_\alpha : H \longrightarrow \{1, -1\}.$$

The characters are nontrivial and pairwise distinct. Indeed, triviality would put β_α in $\overline{L}_n$, making α/a^3 a square. Equality of two characters would put $\beta_\alpha/\beta_\gamma$ in $\overline{L}_n$, making their square classes equal. Both conclusions are excluded by Proposition 5.2.

The Artin–Schreier image is stable under H , so the quotient $V = \overline{M}_n/\wp(\overline{M}_n)$ is an $\mathbb{F}_3[H]$ -module. Its dimension need not be finite. For any character χ of H , the finite group-algebra operator

$$P_\chi = |H|^{-1} \sum_{h \in H} \chi(h)^{-1} h$$

is defined over $\mathbb{F}_3$, since $|H|$ is a power of two. Character orthogonality gives $P_{\chi_\alpha}[\beta_\gamma] = \delta_{\alpha,\gamma}[\beta_\gamma]$. Apply this operator to any proposed finite linear relation among the $[\beta_\gamma]$. Lemma 6.1 says that the selected class is nonzero, so its coefficient must vanish. Doing this for every α proves independence.

For each parent α , Lemma 3.1 gives its three children from β_α, z_α . Thus the right side of (19) contains all roots at height $n+1$. Conversely, their Vandermondes recover the quadratic radicals, and the differences in (7) recover each z_α . The right side is exactly the splitting field. The degree formula now follows from Lemma 3.3 and the proved rank 3^n . $\square$

6.3 Only one local class

The preceding character projections are global. There is no reason to expect that they preserve one chosen place, and no such claim is used. At a chosen completion the direct comparison of principal parts gives a different answer.

Proposition 6.3 (Local rank and the next infinity index). *Under (17), the images of all $[\beta_\alpha]$ in each completion of $\overline{M}_n$ above infinity span exactly one dimension over $\mathbb{F}_3$. The corresponding local extension to $\overline{L}_{n+1}$ is totally ramified of degree three. Consequently*

$$e_\infty(\overline{L}_{n+1}/\overline{k}(t)) = 2 \cdot 3^{n+1}. \quad (20)$$

Proof. Fix such a completion K . By Lemma 6.1, every root has valuation -2 . For two roots α, γ , the equations $f^{\circ n}(\alpha) = f^{\circ n}(\gamma) = t$ and monicity show that the residue of the unit α/γ has 3^n th power one. In characteristic three, $X^{3^n} - 1 = (X - 1)^{3^n}$, so this residue is one.

It follows that the residue of $\beta_\alpha/\beta_\gamma$ is either 1 or -1 . Both square roots have pole order one. For a suitable $\varepsilon \in \{1, -1\} \subset \mathbb{F}_3$, their principal parts therefore cancel:

$$\beta_\alpha - \varepsilon\beta_\gamma \in \mathcal{O}_K.$$

Every integral element of K is an Artin–Schreier image by Lemma 3.4. Hence $[\beta_\alpha] = \varepsilon[\beta_\gamma]$ in $K/\wp(K)$. At least one, and in fact every one, is nonzero because its pole order is one. The span has dimension exactly one.

Formula (19) and Lemma 3.3 now give local degree three. The pole-order calculation in Lemma 3.4 makes this extension totally ramified. The preceding Kummer step is split, so multiplication of ramification indices gives (20). $\square$

6.4 Completing the induction

The last two propositions establish different conclusions: one gives the degree of the global composition, and the other gives the degree of a single local extension. Neither rank is inferred from the other. Combining Proposition 5.2 with Proposition 6.2 yields

$$[\overline{L}_{n+1} : \overline{L}_n] = 2^{2 \cdot 3^{n-1}} 3^{3^n} = \frac{|E_{n+1}|}{|E_n|}.$$

The first induction hypothesis and the upper bound $\text{Gal}(\overline{L}_{n+1}/\overline{k}(t)) \subseteq E_{n+1}$ therefore force equality at height $n + 1$. Proposition 6.3 establishes the second induction hypothesis at that height. Both base cases were proved in Lemma 3.2, so the simultaneous induction is complete for every $n \geq 1$.

Proposition 5.2 and the two Artin–Schreier propositions also prove every assertion of Theorem 2.2, including complete Kummer splitting at infinity and the precise local rank. The infinity-index assertion of Theorem 2.3 has been proved without using a different or genus calculation.

6.5 Descent to the original field

Return to an arbitrary field k of characteristic three. The signature upper bound, with its compatible labelling, was proved over k . The geometric calculation just completed gives

$$|E_n| = [\overline{L}_n : \overline{k}(t)] \leq [L_n : k(t)] \leq |E_n|.$$

All inequalities are equalities. The natural restriction of the geometric group into the arithmetic tree action is therefore an isomorphism, and both groups are exactly E_n in the chosen labelling.

Equality of the degrees also says that extension of constants to $\bar{k}$ does not split the function field: the natural map

$$L_n \otimes_{k(t)} \bar{k}(t) \longrightarrow \bar{k}L_n$$

is an isomorphism. Indeed, it is a surjective homomorphism of $\bar{k}(t)$ -algebras of the same finite dimension. The tensor product is thus a field. Since $L_n/k(t)$ is separable, this is geometric integrality over k , equivalently regularity of L_n/k . In particular no nontrivial algebraic constant extension of k is contained in L_n . This argument does not require k to be perfect.

The labelling was chosen once on the infinite tree, so these finite-height equalities respect restriction. Passing to inverse limits proves the last assertion of Theorem 2.1.

7 The different and the geometric genus

We continue to use geometric constants for all local calculations. The zero-place index was determined without the group induction in Lemma 5.1. The extension there is tame, so

$$d_{0,n} = e_{0,n} - 1 = 2^n - 1.$$

At infinity, it is simpler to calculate the different first in one root field and then pass to the Galois completion.

Proposition 7.1 (Infinity different). *The different exponent of $\bar{L}_n/\bar{k}(t)$ at every place above infinity is $3^{n+1} - 2$.*

Proof. Choose a root X with $t = f^{\circ n}(X)$. Its function field is the rational field $\bar{k}(X)$. With $u = 1/X$, the local reciprocal map at infinity is

$$h(u) = \frac{1}{f(1/u)} = \frac{u^3}{1 + au}, \quad h'(u) = \frac{-au^3}{(1 + au)^2}.$$

Thus $\text{ord}_u h = 3$ and $\text{ord}_u h' = 3$. The base uniformizer is $t^{-1} = h^{\circ n}(u)$. The root completion $\bar{k}((u))/\bar{k}((t^{-1}))$ is separable, totally ramified of degree 3^n . The derivative chain rule gives

$$\begin{aligned} \text{ord}_u((h^{\circ n})'(u)) &= \sum_{j=0}^{n-1} \text{ord}_u h'(h^{\circ j}(u)) \\ &= 3 \sum_{j=0}^{n-1} 3^j = \frac{3(3^n - 1)}{2}. \end{aligned} \tag{21}$$

By the local different formula, this is the different exponent of the root completion over the base completion.

Choose a place of $\bar{L}_n$ extending this root-field place. Its ramification index over the base is $2 \cdot 3^n$, already proved in Section 6. All residue fields are $\bar{k}$, so the Galois completion has degree two over the root completion. That relative extension is tame, with different exponent one. Transitivity (12) gives

$$d_{\infty,n} = 1 + 2 \cdot \frac{3(3^n - 1)}{2} = 3^{n+1} - 2.$$

All places over infinity have the same data because the global extension is Galois. $\square$

We can now complete Theorem 2.3. The ramification indices at both zero and infinity exceed one, so these two points are genuinely branched. Lemma 5.1 excludes any other branch values. Put $N_n = |E_n|$. Over algebraically closed constants there are $N_n/e_{0,n}$ places over zero and $N_n/e_{\infty,n}$ places over infinity. The Riemann–Hurwitz formula for the separable Galois cover therefore reads

$$\begin{aligned} 2g_n - 2 &= N_n \left(-2 + \frac{d_{0,n}}{e_{0,n}} + \frac{d_{\infty,n}}{e_{\infty,n}} \right) \\ &= N_n \left[-2 + (1 - 2^{-n}) + \left(\frac{3}{2} - 3^{-n} \right) \right] \\ &= N_n \left(\frac{1}{2} - 2^{-n} - 3^{-n} \right). \end{aligned}$$

This is (6), and all assertions of Theorem 2.3 follow.

For orientation, the first two instances of these proved formulas are displayed in Table 1. The table is a specialization of the all-height proof, not evidence from which the theorem is inferred.

Table 1: The first two geometric covers, obtained from the formulas.

n	$ E_n $	$e_{0,n}$	$d_{0,n}$	$e_{\infty,n}$	$d_{\infty,n}$	g_n
1	6	2	1	6	7	0
2	648	4	3	18	25	46

The local/global distinction is already visible in the step from height one to height two. The global quadratic and additive degrees are 4 and 27, so the global degree is multiplied by 108. At infinity the quadratic step is split and the additive degree is only 3, so the local index is multiplied by 3. The general induction proves exactly this disparity with 3^n global additive classes at height n and one local class.

8 Conclusion and scope

The cubic $X^3 + aX^2$ realizes the classical group E over every characteristic-three field, without enlarging the constant field. Its geometric ramification is determined at all heights by the same induction that proves maximality of the tree group. The zero-place parity vectors detect exactly the sibling square relations. Distinct Kummer characters then separate all global Artin–Schreier classes, while cancellation of their principal parts makes the local class space at infinity one-dimensional. The different and genus follow from this local information by standard formulas.

The theorem is specific about its domain. The parameter a is nonzero, the characteristic is three, and the target t is transcendental. Specializing t , changing the characteristic, or replacing the inverse-image height by forward iteration time requires separate arguments. In particular, the result supplies no formula for ordinary periodic points and no assertion that a finite specialization has the full generic Galois group. The geometric branch data likewise make no residue-degree assertion over a non-algebraically-closed constant field.

Finally, an inverse-image Galois tower and its genera are source arithmetic data. No target Euler factors, functional equation, root number, correspondence with zeros, or Hilbert–Pólya operator is constructed here. The conclusion is the explicit wild tower and the simultaneous global/local rank mechanism proved above.

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